THE CLOSED SUBSCHEME CYCLE CONSTRUCTION IS COMPATIBLE WITH BASE CHANGE, RESTRICTION AND FLAT PULLBACK

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Allowed standard prerequisites (editorial).

- Cycle family associated to a closed subscheme (relative-cycles chapter, Example example-family-associated-closed, canonical tag 0H4S). cited at statement line 756. The construction whose compatibility with the three operations is asserted.
- Cycle family associated to a coherent module (relative-cycles chapter, Lemma lemma-family-associated-module, canonical tag 0H4R). cited at proof lines 761-762. Supplies the compatibility with base change, restriction and flat pullback for the module construction.
- Cycle of a closed subscheme equals the cycle of its structure sheaf (Chow Homology, Lemma chow-lemma-cycle-closed-coherent, canonical tag 02QY). cited at proof line 763. Sews the closed subscheme construction to the module construction with F = (Z -> X)_* O_Z.

Source reference. Original reference 0H4S: sources/stacks-chow/tags.txt, source lines 21362–21362. Cited at statement line 756: the construction being asserted compatible.

Source reference. Original lemma 0H4R: sources/stacks-chow/tags.txt, source lines 21361–21361. Cited at proof lines 761-762: supplies the three compatibilities.

Source reference. Original lemma 02QY: sources/stacks-chow/tags.txt, source lines 3605–3605. Cited at proof line 763: sews the closed construction to the module construction.

Lemma 0.2. *The construction in Example 0H4S is compatible with base change, restriction, and flat pullback.*

Proof. Taking $\mathcal{F} = (Z \to X)_*\mathcal{O}_Z$ this is a special case of Lemma 0H4R. See Chow Homology, Lemma 02QY. □

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